

Omar Soliman

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EDUCATION

Western University

B.S. in Computer Science, Dean's Honour List

London, Ontario

Expected May 2028

EXPERIENCE

Gestalt Communications

AI Engineer Intern

Oakville, Ontario

July 2026 – Present

- Built a pipeline ingesting United Nations procurement notices and awards to surface bidding opportunities relevant to Gestalt's services; backfilled **280k+ records** on a rate limited PostgreSQL job queue.
- Extended it with a document pipeline: attachment retrieval, PDF/DOCX extraction, **Document AI OCR fallback** for scanned files, sentence aware chunking, and Vertex AI embeddings stored in pgvector.
- Persisted document text and chunks in **transactional checkpoints** before embedding, allowing failed embedding stages to resume without repeating extraction or chunking.
- Provisioned **Terraform managed GCP** infrastructure spanning Cloud Run Jobs, Cloud SQL, a versioned GCS archive, workload scoped IAM, Secret Manager, and monitoring alerts.
- Configured CI with **PostgreSQL integration tests** covering concurrent job claims, expired lease recovery, and duplicate free record versioning, alongside lint and type checks.

Swish Solar

Software Engineer Intern

Waterloo, Ontario

Sept. 2025 – May 2026

- **Sole engineer** on a Python pipeline converting solar SCADA and weather exports into monthly performance and data health PDFs, built to replace manual reporting.
- Built **Isolation Forest** anomaly detection on interval performance ratio, power residual, and AC/DC efficiency features rather than raw power output, to reduce false positives from normal sunrise and sunset dips.
- Added rule based fault categories, severity, and recommended actions so operators know what to act on; grouped consecutive **zero output daylight intervals** into events for reporting.

PROJECTS

Agent Firewall

Python, FastAPI, MCP, Claude API

- Owned the **reasoning engine** of an MCP tool-call proxy that allows, flags for human review, or blocks tool calls.
- Combined regex, Claude semantic scoring, and source trust weighting; assessed **tool call drift** against session intent captured before untrusted content arrived.
- Ran intent extraction and injection detection concurrently with **asyncio** under per stage timeouts; generated reviewer explanations only for non allow decisions.
- Evaluated 504 calls across 146 scenarios: **87% scenario clustered recall** at a 12% false positive rate, versus 50% recall at 13% for a regex baseline.

Microbiome IBD Classifier

Python, scikit-learn, XGBoost, FastAPI, React

- Compared XGBoost and random forest models with **L1 logistic feature selection** for Crohn's disease versus ulcerative colitis classification using 2,105 IBD samples and 2,893 species level features.
- Traced an inflated 0.91 ROC-AUC to study overlap across splits and test set exposure during feature selection and tuning; rebuilt with **cohort grouped cross validation** and fold specific L1 selection, reporting a corrected 0.64.

SMS/MMS Campaign Tool

Python, SQLite, Twilio

- Built and operated a campaign CLI for a municipal mayoral campaign with **delivery status reconciliation**, STOP/START opt-out handling, and SQLite backed resumable sends keyed by campaign, contact, and channel.
- Sent **5k+ MMS messages** in two waves for voter outreach and volunteer recruiting, driving 55 lawn sign opt-ins.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, JavaScript, Bash

Frameworks & Libraries: FastAPI, React, Pydantic, asyncio, Playwright, pandas, NumPy, seaborn

AI & Machine Learning: Claude API, Vertex AI Embeddings, Google Document AI, scikit-learn

Cloud & Tools: GCP, PostgreSQL, MySQL, pgvector, Supabase, Terraform, Docker, Git, GitHub Actions, Linux